open universe, 75, 76 (fig.)
overview effect, 7

parallax, in distance measurement, 117, 118, 119 (fig.)
Parkes radio telescope, 24
particle colliders, 48. *See also specific colliders*
  early universe research using, 134–35
  gravity theory experiments and, 161
  particle physics research using, 135–36
  public concerns about, 129, 130–31, 132, 133
  quark-gluon plasma re-creation in, 44
  safety of, 129
  vacuum decay and, 152
particle horizon, 82–84
particle physics, 11
  Grand Unification in, 37
  gravity and, 160, 164
  particle collider experiments for insights into, 135–36
  symmetry in, 138
Pauli exclusion principle, 122
Peebles, Jim, 23, 24–25, 27n
Peiris, Hiranya, 179, 180, 187, 202, 206
Penrose, Roger, 175, 208
Penzias, Arno, 23–24, 25–27
periodic table of the elements, symmetry in, 137
phantom dark energy, 128
  Big Rip and, 115
  Caldwell's calculation of, 110–12, 111 (fig.), 115
phenomenology, 11–12
physics
  areas studied by cosmologists in, 10
  Grand Unified Theory in, 36–37
  gravity theory and, 161
  Higgs field and, 134
  models in, 22n
  new perspectives on the universe's end and findings in, 7–8
  particle collider experiments for insights into, 135, 141